Write a new value under an old key. Start from S_2 of the interference figure, where $k_1 = (1, 0)$ holds $v_1 = (3, 1)$ and $k_2 = (0, 1)$ holds $v_2 = (-1, 2)$. The third write is (k_3, v_3) with $k_3 = k_1$ and $v_3 = (0, 4)$.

Additive rule (linear attention)

$S_3 = S_2 + k_3 v_3^\top =$ $\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ $+$ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\begin{bmatrix} 0 & 4 \end{bmatrix}$ $=$ $\begin{bmatrix} 3 & 5 \\ -1 & 2 \end{bmatrix}$

$k_1^\top S_3 =$ $\begin{bmatrix} 3 & 5 \end{bmatrix}$ $= v_3^\top +$ $\begin{bmatrix} 3 & 1 \end{bmatrix}$ the old value v_1 stays

$k_2^\top S_3 =$ $\begin{bmatrix} -1 & 2 \end{bmatrix}$ $= v_2^\top$ ✓

Delta rule, with $\beta_3 = 1$

1. Read

$k_3^\top S_2 =$ $\begin{bmatrix} 3 & 1 \end{bmatrix}$

what k_3 reads now

2. Correct

$u_3^\top = v_3^\top - k_3^\top S_2 =$ $\begin{bmatrix} 0 & 4 \end{bmatrix}$ $-$ $\begin{bmatrix} 3 & 1 \end{bmatrix}$ $=$ $\begin{bmatrix} -3 & 3 \end{bmatrix}$

only the difference is written: the pseudo-value u_3

3. Write

$S_3 = S_2 + k_3 u_3^\top =$ $\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ $+$ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\begin{bmatrix} -3 & 3 \end{bmatrix}$ $=$ $\begin{bmatrix} 0 & 4 \\ -1 & 2 \end{bmatrix}$

$k_1^\top S_3 =$ $\begin{bmatrix} 0 & 4 \end{bmatrix}$ $= v_3^\top$ ✓

$k_2^\top S_3 =$ $\begin{bmatrix} -1 & 2 \end{bmatrix}$ $= v_2^\top$ ✓

With a key of unit length, the write leaves $S_t^\top k_t = (1 - \beta_t)\, S_{t-1}^\top k_t + \beta_t\, v_t$, a mix of the old read and the new value: $\beta_t = 1$ replaces it, $\beta_t = 0$ keeps it.